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Activity History: Debian Linux system Installation and Optimization

Developer: HMP

I. Project Overview

This document summarizes a technical session focused on Debian Linux system installation, configuration, and optimization.

SCOPE OF WORK

- Linux installation using Debian
- Manual partitioning
- Post-install system configuration and tuning

II. System Installation and Storage Architecture

A Linux installation was performed with manual configuration steps.

Disk layout included an EFI system partition and a separate /boot partition.

The remaining disk space was configured with LVM logical volume management for the root filesystem.

STORAGE DESIGN

- EFI system partition (~500MB)
- Root partition (~1GB)
- Physical volume
- LVM volume group with root logical volume

III. System Configuration and Package Management

Post-install configuration included repository definition, package management adjustments, and system optimization settings.

APT configuration was modified to reduce automatic installation of recommended and suggested packages.

System packages were audited and selectively removed to reduce installed footprint.

CONFIGURATION ACTIONS

- APT repository configuration
- Disabling of recommended/suggested package installation
- Package audit and removal of non-essential components
- System service review and selective disabling

IV. Security, Networking, and Performance Adjustments

Networking configuration was migrated to NetworkManager, replacing legacy interface-based configuration.

Keyring management was configured to support session-based credential unlocking.

Performance-related adjustments were applied including swappiness tuning and journald log size limitation.

SYSTEM ADJUSTMENTS

- NetworkManager-based networking configuration
- GNOME keyring integration via PAM
- Systemd journal size limitation
- Swappiness set for reduced swap usage